UNIT - II

Conceptual Design & Introduction to Earthquake Resistant Design

Syllabus (as given):

Functional planning - continuous load path - overall form - simplicity & symmetry - elongated shapes - stiffness & strength - horizontal & vertical members - twisting - ductility - flexible buildings - framing systems - choice of materials (unconfined/confined concrete, masonry, steel). Seismic design requirements - regular/irregular configurations - assumptions - design loads - load combinations - permissible stresses - methods of analysis - factors in analysis - equivalent lateral force method.

PART A : CONCEPTUAL DESIGN

1.1 Introduction

Conceptual (architectural + structural) design decides how a building will behave in an earthquake before any calculation is done. Good configuration is cheaper and more reliable than heavy detailing on a poor configuration. "A good building is made, not analysed" - failures in Bhuj 2001, Kobe 1995, Northridge 1994 were mostly due to bad configuration: soft storeys, torsion, short columns, pounding.

Definition - Seismic design philosophy (IS 1893)

Structure should: (i) resist minor shaking (frequent) without damage; (ii) resist moderate shaking with limited structural damage and some non-structural damage (repairable); (iii) resist major (MCE) shaking without collapse - life safety, with damage allowed. Achieved by strength + ductility (reducing the design force by factor R).

- Why not design elastic for MCE? Elastic design force would be R times larger (3-5x) -> uneconomical. Instead allow inelastic energy dissipation.
- Four virtues of EQ-resistant buildings (Arnold & Reitherman): Good structural configuration, adequate lateral stiffness, adequate lateral strength, good ductility.
- Relevant codes: IS 1893 (Part 1):2016 (design loads), IS 13920:2016 (ductile detailing of RC), IS 4326:2013 (low-rise buildings), IS 13828 (low-strength masonry), IS 13827 (earthen), IS 15988 (retrofit of RC).

1.2 Functional Planning

- Plan the layout of rooms, stairs, lifts, services so that structural elements can be arranged symmetrically and uniformly - do not let architecture demand floating columns, large openings in floors, discontinuous walls.
- Locate heavy masses (water tanks, storage, plant rooms) low and near the centre; avoid heavy roof loads (reduces base shear since $V = A_h W$).

- Stair/lift cores should be symmetrically located or isolated from the main frame (otherwise they stiffen one side -> torsion).
- Provide seismic joints (separation gaps) where a building has sudden changes in shape, so each part vibrates independently; also to avoid pounding with adjacent buildings.
- Keep ground storey open car-parking (stilt) only if the frame is specially stiffened - otherwise it becomes a soft storey.
- Keep spans moderate, storey heights uniform, light non-structural partitions.

> Seismic gap
> Gap for pounding (IS 1893:2016 cl. 7.11.1.3): separation = R x design lateral displacement of each adjacent unit (sum of the two); typically 25-40 mm for low-rise. Floors of adjacent buildings should ideally be at the same level.

1.3 Continuous Load Path

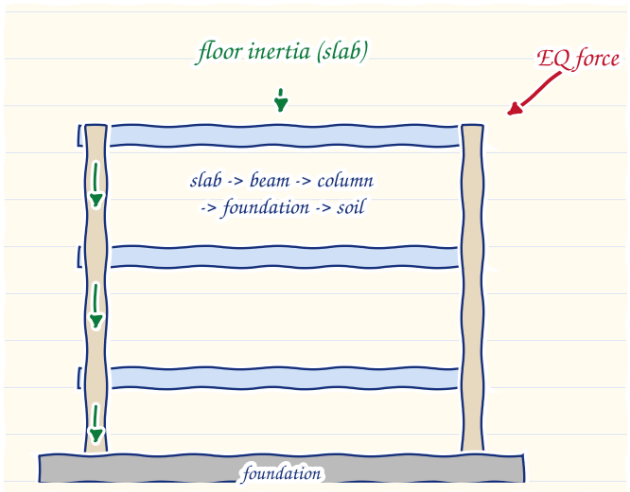


Fig: Load path for gravity and lateral (earthquake) forces

- A continuous, direct load path must carry inertia forces from every mass (slab, walls, roof) -> diaphragm -> vertical elements (columns/shear walls) -> foundation -> soil. Any break (weak link) is where failure starts.
- Floor slabs act as rigid diaphragms distributing lateral loads to vertical resisting elements by stiffness.
- Connections: beam-column joints, roof-to-wall, wall-to-foundation, wall-to-wall (corner) must be at least as strong as members connected.
- Walls/columns should run vertically down to the foundation without discontinuity (avoid floating columns, hanging walls, discontinuous shear walls).

1.4 Overall Form - Simplicity and Symmetry

- Simple, compact, symmetrical plan in both axes (square, rectangle, circle) gives uniform stress, minimal torsion, easy to analyse and detail. Symmetry applies to both geometry (mass) and stiffness.
- Centre of mass (CM) and centre of rigidity/stiffness (CR) should coincide: eccentricity e = |CM - CR| as small as possible. Torsional moment = V x e.

- Complex shapes (L, T, U, H, Y, +, E) develop stress concentration at re-entrant corners and different wings vibrate with different periods -> separate them by seismic joints into simple rectangles.
- Vertical form: continuous, uniform elevation; avoid abrupt set-backs, overhangs (cantilevers), heavy upper floors, tapered mass at top.

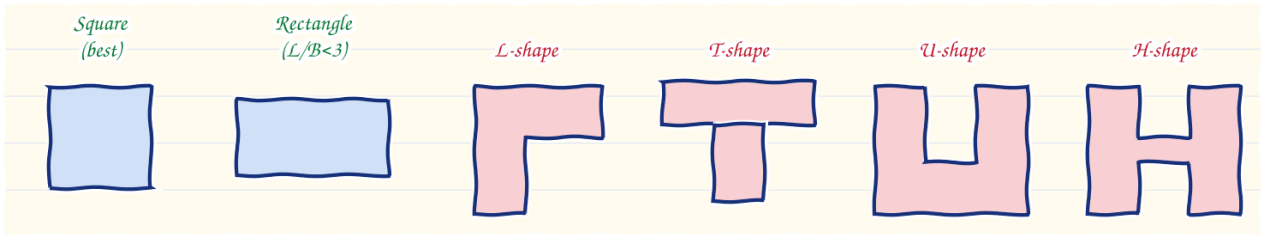


Fig: Regular (blue/good) vs irregular plan shapes (pink/poor)

Elongated shapes

- Long buildings (L/B > 3) suffer because ground motion arrives with phase difference at the ends; long floor diaphragms behave flexibly and are subjected to differential motion -> tension/cracking in the diaphragm, large torsion.
- Limit plan aspect ratio L/B ≤ 3 (IS 4326 limits to ≤ 3 for masonry); otherwise provide seismic joints at suitable intervals.
- Narrow tall buildings (H/B > 4) -> overturning, large axial force changes in end columns; keep slenderness H/B small.

1.5 Stiffness and Strength; Horizontal & Vertical Members

- Stiffness (lateral) controls period T, storey drift and non-structural damage. Stiffness too small -> large drift (P-delta, damage); too large -> attracts higher force (A_h increases as T decreases up to the plateau).
- Strength controls whether the structure yields. Strength must be uniform in plan and elevation - avoid weak storey (strength < 80% of storey above).
- Vertical members (columns, shear walls) carry gravity and resist lateral loads; they must be strong and ductile - "strong column - weak beam" so that yielding occurs in beams (energy dissipated in many places) and not in columns (storey collapse).
- Horizontal members (beams, slabs/diaphragms) transfer inertia forces to vertical members; beam failure is less catastrophic than column failure. Keep slab continuous (opening < 50% of area), well connected to walls/columns.
- Use shear walls/bracing at periphery to give maximum torsional resistance.

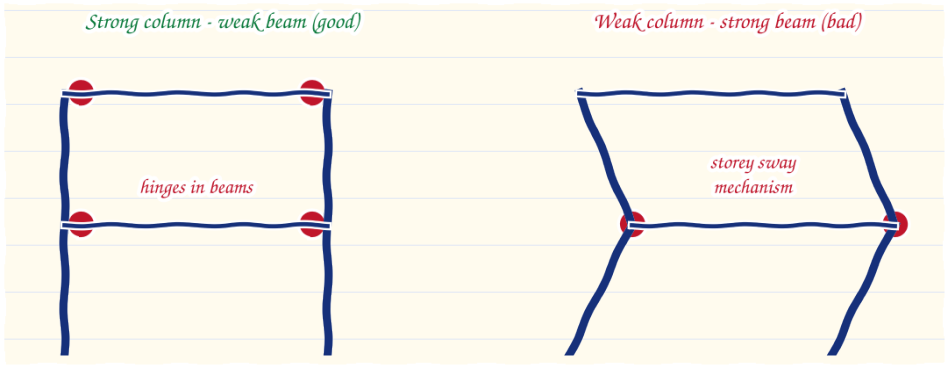


Fig: Beam-sway vs storey mechanism

1.6 Twisting of Buildings (Torsion)

- Arises when CM ≠ CR (mass or stiffness unsymmetrical), or from rotational component of ground motion -> building rotates about CR, columns far from CR undergo extra displacement.
- Torsion causes high demand in corner/edge columns, large differential drift and often collapse (e.g., corner buildings with ground floor stiffness only on two sides).
- IS 1893: design eccentricity $e_d = 1.5\, e_s + 0.05\, b$ or $e_s - 0.05\, b$ (more severe) where e_s = static eccentricity, b = plan dimension perpendicular to force.
- Remedies: symmetric layout; place stiff elements (walls, cores) at the periphery, symmetrically; avoid core placed at one end; increase torsional stiffness; keep mass symmetric.
- Torsional irregularity: max storey drift at one end > 1.5 x min drift (at the other end) (IS 1893:2016).

1.7 Ductility

Definition - Ductility (μ)
The ability of a structure/member to undergo large inelastic deformation without significant loss of strength, thereby dissipating energy. Displacement ductility $\mu = \Delta_u / \Delta_y$.

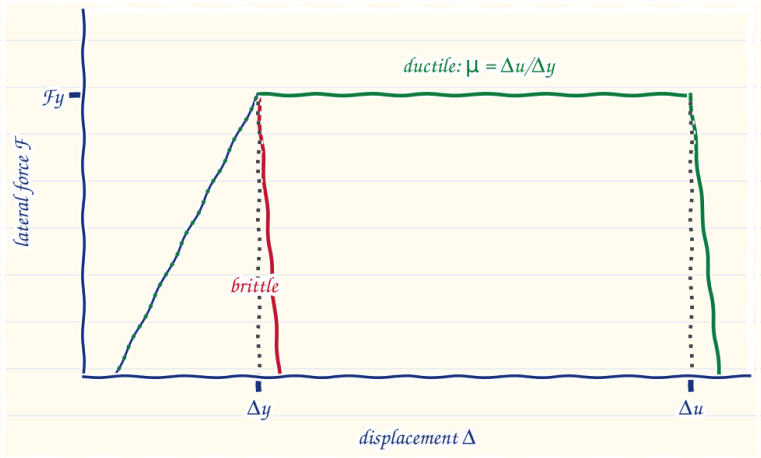


Fig: Ductile vs brittle response

- Ductility relationships: displacement $\mu_\Delta = \Delta_u / \Delta_y$; curvature $\mu_\varphi = \varphi_u / \varphi_y$; rotational $\mu_\theta = \theta_u / \theta_y$; Relationship: $\mu_\Delta = 1 + 3(\mu_\varphi - 1)(L_p/L)(1 - 0.5\, L_p/L)$ (L_p = plastic hinge length).
- Response reduction R and ductility: equal displacement rule (long period): $R_\mu = \mu$; equal energy rule (short period, T < ~0.5 s): $R_\mu = \sqrt{(2\mu - 1)}$.

- Ductility is provided by: closely spaced ties/confinement, adequate anchorage, controlled steel ratio (under-reinforced), tension-controlled sections, good detailing (IS 13920).
- Brittle failures to avoid: shear failure, bond/anchorage failure, crushing of concrete, buckling of bars, joint failure, masonry diagonal tension.
- Capacity design: design the intended yielding parts (beams) first, then make the rest (columns, joints, shear capacity) stronger so that ductile mechanism forms.

1.8 Flexible Buildings

- Flexible (tall, slender, moment frame) buildings have long natural period ($T \approx 1$-4 s) -> attract lower design seismic force (S_d/g falls with T) but suffer larger displacements and drift.
- Problems: large inter-storey drift damaging partitions and cladding, P-delta effects, instability, pounding with neighbours; resonance with long-period ground motion on soft soils.
- Stiff (low-rise, walled) buildings: short period, higher force, small drift, but less ductile and may be brittle (masonry).
- Balanced design: limit drift to 0.004 h under design EQ (IS 1893:2016), keep adequate stiffness using shear walls, bracing; $T \leq 4$ s ideal.

1.9 Framing Systems

System	Features	Remarks
Moment-resisting frame (MRF)	Beams & columns with rigid joints resist lateral loads by flexure	Architecturally flexible; ductile if SMRF ($R = 5$); drift is main issue; best up to ~10-15 storeys
Braced frame (concentric/eccentric)	Diagonal members carry lateral load axially (steel)	Very stiff; EBF more ductile (link beams)
Shear wall system	RC walls resist lateral loads by cantilever action	Stiff and strong; reduces drift; ductile shear wall $R = 4$
Dual system (wall-frame)	Frame + walls share lateral load (frame ≥ 25% of base shear)	Best performance for 10-40 storeys; R = 4.5-5
Tube / core-outrigger	Perimeter closely spaced columns acting as tube	Tall buildings > 40 storeys
Infilled frame	Masonry infill within frame	Stiffens frame; but short-column/soft-storey risk, brittle

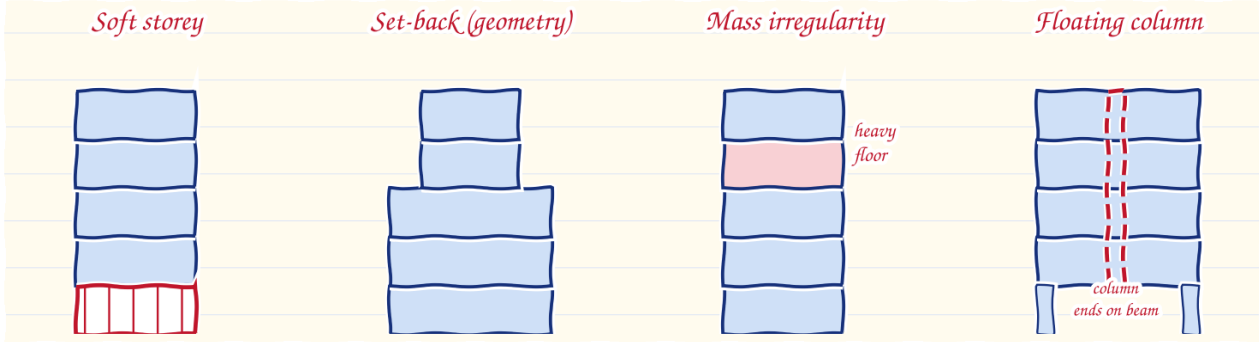


Fig: Vertical irregularities: soft storey, set-back, mass, floating column

1.10 Choice of Construction Materials

Material	Seismic behaviour
Structural steel	Most ductile (yield plateau, elongation 20%+); light weight; high strength/weight -> lower inertia; but local/lateral buckling & connection (weld) failures (Northridge) need care
Reinforced concrete	Good if detailed ductilely (IS 13920); cheap, durable; heavy; cover spall and brittle shear if poorly confined
Masonry	Brittle, low tension/shear strength, heavy; unreinforced masonry is the most vulnerable (needs bands, vertical steel - IS 4326)
Timber	Light, ductile connections; good seismic performance
Earth / adobe	Heavy, weak; need reinforcement or confinement

(a) Unconfined concrete

- Plain concrete is strong in compression but brittle: stress-strain curve rises to f_c at strain ≈ 0.002 and falls quickly; crushing strain ε_{cu} ≈ 0.0035 (IS 456).
- Cover concrete spalls at the end of the strength; sudden loss of capacity.

(b) Confined concrete

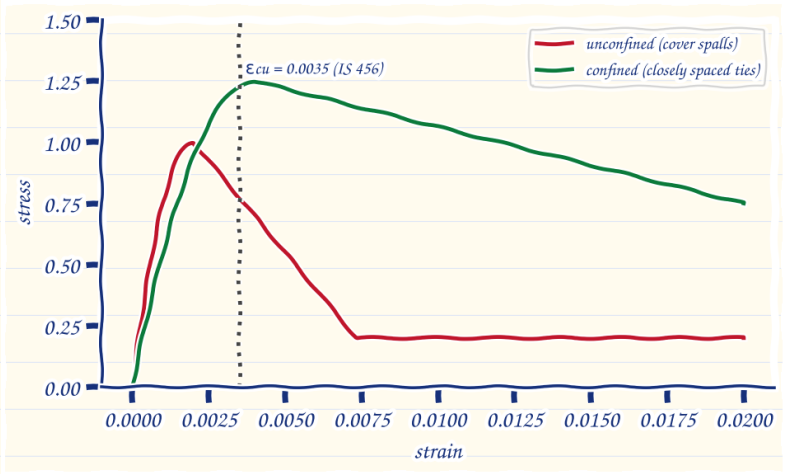


Fig: Confinement increases strength and ductility

- Closely spaced transverse ties/spirals restrain lateral expansion of core -> triaxial compression -> higher strength f_{cc} and much larger ultimate strain (0.01-0.025) - slower descending branch.
- Effectiveness increases with smaller spacing, larger volumetric ratio ρ_s, 135° hooks, circular hoops (best), and supplementary cross-ties.
- Confinement also prevents buckling of longitudinal bars and gives shear strength.

(c) Masonry and (d) Reinforcing steel

- Masonry: good in compression only; tension strength ≈ 0; shear strength depends on mortar and vertical load. Use richer mortar (1:4 or 1:6 cement-sand), strong units (brick ≥ 3.5 MPa), reinforcement & bands.
- Reinforcing steel (IS 13920:2016): Fe 415, Fe 500, Fe 550 (and D grades with better ductility); elongation ≥ 14.5%, actual yield strength not more than 20 N/mm² above nominal, UTS/YS ≥ 1.15 -> ensures ductile yielding and strain hardening. Prefer TMT bars; avoid cold-twisted (CTD) bars.

- Minimum concrete grade: M20 (M25 for buildings > 15 m height in Zones III, IV, V under IS 13920:2016). Avoid very high-strength concrete for ductility reasons (more brittle).

PART B : INTRODUCTION TO EARTHQUAKE RESISTANT D

2.1 Seismic Design Requirements

- Strength: members and connections to resist design forces (limit state design with factored loads, IS 456).
- Stiffness: storey drift ≤ 0.004 h (IS 1893:2016) under design lateral force (DBE, partial load factors 1.0).
- Ductility: IS 13920 detailing in Zones III-V (and Zone II for important buildings).
- Stability: overturning, sliding, P-delta, foundation (liquefaction, differential settlement).
- Integrity: all parts tied together; non-structural elements anchored.
- Both horizontal & vertical (for long spans, cantilevers, prestressed members) shaking considered; torsion included.

2.2 Regular and Irregular Configurations (IS 1893 (Part 1):2016, Cl. 7.1)

Regular building = simple, symmetric, uniform in plan and elevation. Irregular = having any of the following irregularities; it must be analysed dynamically if taller than limits, and R may be reduced / extra checks applied.

Plan irregularity (Table 5)	Limit
Torsional irregularity	Max drift at one end > 1.5 x average / min drift of the two ends
Re-entrant corners	Projection beyond re-entrant corner > 20% of plan dimension in that direction
Diaphragm discontinuity	Openings/cut-outs > 50% of gross floor area, or sudden change in stiffness > 50%
Out-of-plane offsets	Discontinuities in lateral force path (offset of vertical elements)
Non-parallel systems	Vertical lateral-force elements not parallel/symmetric to major orthogonal axes

Vertical irregularity (Table 6)	Limit
Stiffness irregularity - soft storey	Lateral stiffness < 70% of storey above or < 80% of average of three storeys above
Mass irregularity	Seismic weight of any storey > 150% of adjacent storey (roof excepted)
Vertical geometric irregularity	Horizontal dimension of lateral system > 125% of adjacent storey
In-plane discontinuity	Offset of lateral-force resisting element > its length
Strength irregularity - weak storey	Storey lateral strength < 80% of the storey above
Floating / stub columns	Columns not carried to foundation; very short columns

2.3 Basic Assumptions (IS 1893)

- Earthquake causes impulsive ground motions, which are complex and irregular; they do not occur simultaneously with maximum wind or maximum flood or maximum sea waves/current.
- Design EQ force is not combined with wind; loads: EQ + dead + part live.
- Earthquake forces are idealised as equivalent static lateral forces at floor levels (for regular, low-mid rise) - dynamic analysis otherwise.
- Modulus of elasticity of materials, wherever required, may be taken as for static analysis, unless a dynamic value is available (RC: $E_c = 5000\sqrt{f_{ck}}$). Floors/diaphragms rigid in plane.
- Soil-structure interaction usually ignored except for special cases; seismic weight W = DL + fraction of IL.

2.4 Design Earthquake Loads - Seismic Weight

Imposed (live) load on floor	Fraction of IL included in seismic weight
Up to 3.0 kN/m²	25%
Above 3.0 kN/m²	50%
Roof	0% (live load on roof not considered)

Design horizontal seismic coefficient (IS 1893:2016 cl. 6.4.2)

$A_h = (Z/2) \times (S_a/g) / (R/I)$

Z = zone factor (0.10, 0.16, 0.24, 0.36); I = importance factor; R = response reduction factor; S_a/g = design spectrum (5% damping)

Design base shear $V_B = A_h W$ (W = seismic weight of building)

Factor	Values
Importance factor I	1.5 - important service & community buildings (hospital, fire station, power, telecom); 1.2 - business & residential > 200 persons (assembly halls, malls); 1.0 - all other
Response reduction R	OMRF 3.0; SMRF 5.0; ordinary RC shear wall 3.0; ductile shear wall 4.0; shear wall + SMRF dual 4.5-5.0; unreinforced masonry 1.5; reinforced masonry 3.0 (as per IS 1893 Table 9)
Soil types (Sa/g)	Type I rock/hard; Type II medium; Type III soft - different plateaus & decay (see spectrum)
Vertical seismic	Vertical acceleration = 2/3 of horizontal design spectrum; needed for cantilevers > 1.5 m, prestressed beams, long spans > 20 m (IS 1893:2016)

Important / Exam point

Time period (empirical, IS 1893:2016 cl 7.6.2): RC moment frame without infill: $T_a = 0.075\ h^{0.75}$; steel: $T_a = 0.085\ h^{0.75}$; RC/steel frame with brick infill (or shear wall building): $T_a = 0.09\ h/\sqrt{d}$ (h = height in m, d = base dimension in m along the considered direction).

2.5 Basic Load Combinations (IS 1893:2016 / IS 456 limit state)

Limit state of collapse (IS 456 / 1893)	Combination
Gravity only	1.5 (DL + IL)

Limit state of collapse (IS 456 / 1893)	Combination
With EQ	1.2 (DL + IL ± EL)
	1.5 (DL ± EL)
Uplift / overturning check	0.9 DL ± 1.5 EL
Earthquake in both directions	EL = ± EL_x ± 0.3 EL_y or ± 0.3 EL_x ± EL_y (+ vertical where needed)

- Working stress method: DL + IL ± EL, DL ± EL with allowable stresses increased by 33.3%.
- In IS 1893 EL includes combination of all three components if required: ±EL_x ± 0.3EL_y ± 0.3EL_z.

2.6 Permissible Stresses

- When WSM is used, permissible stresses in materials (concrete, steel, masonry) and allowable soil bearing pressure may be increased by 33.33% for EQ load combinations (IS 1893/IS 456 Annex B).
- For limit state design, partial safety factors: γ_c = 1.5 (concrete), γ_s = 1.15 (steel) generally, but for seismic combos γ_c = 1.5 and γ_s = 1.15 are retained while load factor is reduced (1.2 / 1.5).
- Soil: for liquefiable soils increase not allowed; for soft soils (N < 10) - see Table of IS 1893 for allowable bearing capacity increase.

2.7 Seismic Methods of Analysis

Method	Basis	When used
Equivalent static (ELF / seismic coefficient)	Single mode; $V_B = A_h W$ distributed with height	Regular buildings: Zone II/III up to 90 m; Zone IV/V up to 40 m (regular); irregular buildings < 12 m (IV, V) / < 40 m (II, III)
Response spectrum method	Modal analysis, modes combined by SRSS/CQC; ≥ 90% mass participation	Regular > limits, all irregular buildings above limits
Time history analysis	Step-by-step integration with accelerograms (linear or non-linear)	Important / very tall / irregular buildings, base-isolated, non-linear studies
Pushover (static non-linear)	Incremental lateral load until collapse mechanism	Performance-based design, retrofit evaluation

Design base shear from dynamic analysis V_B (dynamic) must be scaled up to the base shear V_B from empirical period of ELF (so that $V_B \geq V_B$).

2.8 Factors in Seismic Analysis

- Z - seismic zone; I - importance; R - ductility/overstrength reduction; S_a/g - soil & period dependent; W - seismic weight; h, d - height and base dimension for T.
- Soil type and foundation (liquefaction, bearing).
- Damping (5% standard; other values via multiplying factor); Torsion (accidental eccentricity 0.05 b); P-Δ effects; Orthogonal effects; Drift limits.
- Redundancy and over-strength: well-distributed lateral system is better.

2.9 Equivalent Lateral Force Method - Procedure

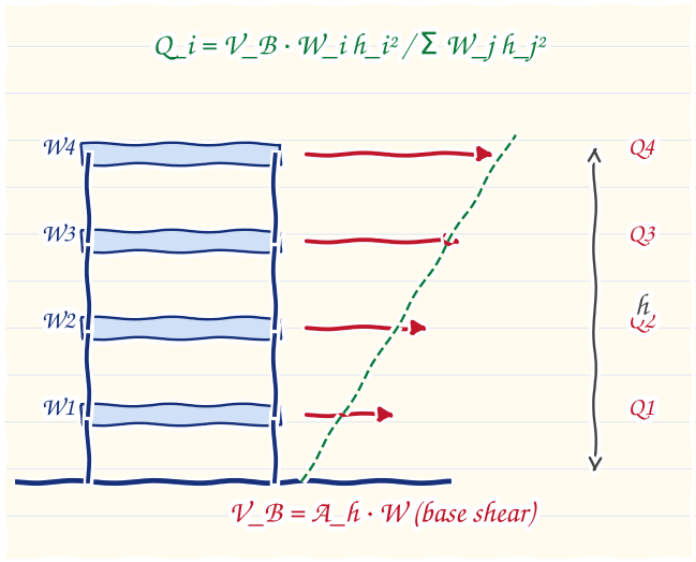


Fig: Equivalent lateral force distribution

1. Find seismic weights W_i of each floor (DL + 25/50% IL; roof IL = 0) and total W.
2. Find fundamental natural period T_a in each direction (empirical formulas above).
3. Read S_a/g for T_a and soil type from the design spectrum (e.g., medium soil, $T > 0.55$: $S_a/g = 1.36/T$, capped 2.5).
4. Compute $A_h = (Z/2)(S_a/g)/(R/I)$ and $V_B = A_h W$.
5. Distribute along height: $Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$ (parabolic distribution).
6. Distribute to lateral-load resisting elements (frames/walls) by stiffness; add torsion; check drift ≤ 0.004 h, P-delta, overturning.
7. Analyse for load combinations, design members, then detail as per IS 13920.

Solved example (ELF)

4-storey RC office building, Zone III ($Z = 0.16$), medium soil, SMRF ($R = 5$), $I = 1.0$; $W = 4 \times 3000$ kN = 12 000 kN; $h = 13$ m; $d = 20$ m; infilled frame.

$T_a = 0.09\, h/\sqrt{d} = 0.09 \times 13/\sqrt{20} = 0.262$ s (< 0.55 s plateau) -> $S_a/g = 2.5$

$A_h = (0.16/2)(2.5)/(5/1) = 0.08 \times 2.5/5 = 0.04$

$V_B = 0.04 \times 12\,000 = 480$ kN (4% of weight)

Continued

Distribution of Q_i (equal floors of 3 m): Q_i proportional to $h_i^2 = 3^2, 6^2, 9^2, 12^2 = 9, 36, 81, 144$ (sum 270): $Q = 480 \times (9, 36, 81, 144)/270 = 16, 64, 144, 256$ kN (check: sum 480 kN).

EXTRA POINTS & NUMERICAL

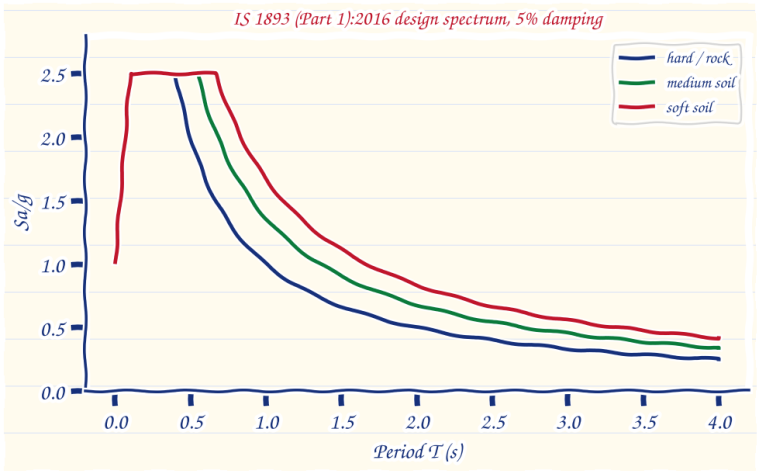


Fig: Design spectrum S_a/g (5% damping) - IS 1893 (Part 1):2016

Soil type	$0 \le T \le 0.10$	Plateau	Decay (up to $T = 4$ s)	$T > 4$ s
Type I - Rock / hard soil	$1 + 15T$	2.5 (0.10-0.40 s)	$1.00/T$	0.25
Type II - Medium soil	$1 + 15T$	2.5 (0.10-0.55 s)	$1.36/T$	0.34
Type III - Soft soil	$1 + 15T$	2.5 (0.10-0.67 s)	$1.67/T$	0.42

Common seismic problems	Why it happens	Remedy
Soft storey	Open ground floor (parking) has less stiffness; all drift concentrates in one storey	Provide RC shear walls or bracing in the open storey; avoid it altogether where possible
Short column	Partial-height infill or mezzanine -> column stiffer than its neighbours attracts more shear -> shear failure	Full-height gap between infill and column, or ductile shear detailing over full height
Pounding	Adjacent buildings of different periods hit each other	Provide seismic gap ($R \times \Delta$)
Torsion	CM ≠ CR	Symmetric layout, peripheral walls
Weak beam-column joint	Shear/anchorage failure at joint	Transverse ties continue through joint (IS 13920)
Heavy parapets/tanks	Large mass at top, whiplash	Reduce mass, anchor, brace

Solved example 2 (hospital, Zone V)

Zone V ($Z = 0.36$), hospital ($I = 1.5$), SMRF ($R = 5$), rock site, $T_a = 0.40$ s, $W = 20\,000$ kN.

$S_d/g = 2.5$ (plateau); $A_h = (0.36/2)(2.5)/(5/1.5) = 0.45/3.333 = 0.135$

$V_B = 0.135 \times 20\,000 = 2700$ kN (13.5% of weight) - for an ordinary building ($I = 1$) it would have been 1800 kN.

Important / Exam point

Difference OMRF vs SMRF: OMRF ($R = 3$) is not detailed per IS 13920 -> allowed only in Zone II; SMRF ($R = 5$) has ductile detailing and is mandatory in Zones III, IV, V. Higher R gives lower design force but demands better detailing.

QUICK REVISION - Likely questions

1. Explain the seismic design philosophy and the four virtues of an earthquake-resistant building.
2. Importance of symmetry, simplicity; effect of plan/elevation irregularities - with sketches.
3. What is ductility? Explain ductility relationships and how it is achieved; confined vs unconfined concrete.
4. Explain torsion in buildings and ways to reduce it; soft-storey and short-column effect.
5. Classify irregularities as per IS 1893 (Part 1):2016; limits.
6. Basic assumptions and load combinations for seismic design; permissible stresses.
7. Explain methods of seismic analysis and conditions of applicability of ELF.
8. Procedure of ELF method; numerical for base shear and its distribution.